

COVID-19, Household Spending Reallocation, and Financial Resilience

I. Introduction

The COVID-19 pandemic represents one of the most significant exogenous shocks to household financial behaviour in modern history. Unlike traditional economic downturns, the pandemic simultaneously disrupted labour markets, constrained consumption opportunities, and heightened economic uncertainty. These forces led to substantial shifts in household expenditure patterns across a wide range of consumption categories. From a financial perspective, these changes are particularly important because household consumption decisions reflect underlying risk preferences, liquidity constraints, and intertemporal optimization behaviour. As such, examining how households adjusted their spending provides valuable insight into financial resilience and the broader transmission of macroeconomic shocks to sectoral demand and asset prices. Figure 1 provides a preliminary illustration of household spending over time, showing a slight decline during the pandemic period, followed by a strong increase in the recovery period. This visual evidence suggests that while aggregate spending remained relatively stable overall, underlying compositional changes may be present, motivating a more detailed empirical analysis.

A growing body of literature examines how households adjust their consumption in response to large economic shocks, and the COVID-19 pandemic provides a uniquely rich empirical setting. Early evidence documents sharp declines in overall consumption, especially in contact-intensive sectors such as travel, dining, and entertainment, reflecting both mobility restrictions and elevated uncertainty (Baker et al. 2020; Chetty et al. 2020). However, subsequent research suggests that these aggregate declines were partially mitigated by government transfers and the accumulation of forced savings, which helped stabilize overall consumption levels during the crisis (Coibion, Gorodnichenko, and Weber 2021; Kaplan, Moll, and Violante 2020).

More recent studies emphasize heterogeneity and compositional changes in consumption behaviour rather than focusing solely on aggregate effects. Andersen et al. (2022) and Parker et al. (2022) demonstrate that households rapidly reallocated expenditures away from services toward goods, reflecting both supply constraints and shifting preferences. These findings highlight the importance of distinguishing between total consumption and its composition when analyzing household responses to economic shocks.

From a theoretical perspective, these empirical patterns are consistent with models of precautionary savings and consumption smoothing. Carroll (1997) develops a buffer-stock saving framework in which households respond to increased income uncertainty by reducing non-essential expenditures while maintaining essential consumption. This framework provides a natural interpretation of pandemic-era behaviour as a form of financial risk management. In addition, Gormsen and Koijen (2021) show that financial markets rapidly incorporated expectations of reduced discretionary consumption into asset prices, particularly for firms with high cyclical demand exposure. Together, these insights suggest that changes in consumption composition may have important implications for financial markets and investment strategies.

This study builds on the existing literature by examining how the pandemic affected spending allocation across consumption categories rather than aggregate consumption alone, and by assessing whether these effects persisted into the recovery period. Specifically, the paper addresses the following research question: how did the COVID-19 pandemic affect household spending allocation across discretionary and non-discretionary categories, and to what extent did these effects reverse during the recovery period? Using data from Statistics Canada's Survey of Household Spending, the analysis employs a difference-in-differences (DiD) framework that exploits variation across expenditure categories, treating discretionary spending as more exposed

to pandemic-related constraints. The empirical design defines 2021 as the treatment period and 2019 as the pre-treatment baseline, while incorporating a recovery indicator for 2023 to evaluate post-pandemic dynamics.

The results indicate that while aggregate spending levels remained relatively stable, there was a statistically significant reallocation away from discretionary spending categories during the pandemic. This pattern is consistent with heightened precautionary behaviour, liquidity constraints, and restrictions on consumption opportunities. Importantly, the recovery period shows only partial reversal of these effects, suggesting that the pandemic induced persistent changes in household spending preferences. These findings contribute to the literature by demonstrating that macroeconomic shocks can generate sustained shifts in consumption composition, with important implications for sectoral investment, asset pricing, and household financial decision-making.

II. Data Selection and Sample

This study relies on publicly available data from Statistics Canada's Survey of Household Spending (SHS), a nationally representative dataset constructed using consistent survey methodologies. While the data are aggregated at the expenditure category level rather than at the individual household level, this level of aggregation is appropriate for the research objective, which focuses on compositional shifts in spending across discretionary and non-discretionary categories rather than household-level heterogeneity. The use of category-level data enables a clear implementation of the difference-in-differences framework by exploiting variation across both categories and time. Table 1 reports summary statistics for the key variables used in the analysis.

The final sample consists of 60 observations for the baseline, difference-in-differences, and fixed-effects specifications, and 90 observations for the recovery models. The analysis focuses on three key periods: 2019, the pre-pandemic baseline; 2021, the pandemic period; and 2023, the recovery period. This structure allows for a clear comparison of spending patterns before, during, and after the pandemic shock.

The empirical analysis is based on several key variables. A pandemic indicator variable is set to 1 for observations in 2021, capturing the immediate effects of the shock. A recovery indicator variable is set to 1 for observations in 2023, enabling the examination of post-pandemic adjustments. A discretionary indicator identifies non-essential expenditure categories, which are expected to be more sensitive to economic uncertainty and mobility constraints. Interaction terms between the pandemic and discretionary indicators, as well as between recovery and discretionary indicators, are included to capture heterogeneous effects across spending types. Table 2 provides detailed definitions of all variables used in the empirical specifications.

The dataset includes category-level expenditures spanning both essential and discretionary consumption, including housing, transportation, and entertainment. While the data are aggregated at the category level rather than at the individual household level, they provide sufficient variation to identify within-category changes over time. This structure is particularly important for implementing difference-in-differences and fixed-effects models, which rely on variation across both dimensions. Although the data are aggregated at the category level, this level of aggregation is appropriate for the research question, which focuses on compositional shifts in spending rather than household-level heterogeneity. The summary statistics in Table 1 further highlight meaningful variation across expenditure categories, supporting the dataset's suitability for the empirical analysis. The correlations reported in Table 3 indicate moderate

relationships among key variables, with no evidence of severe multicollinearity. In particular, the relatively low correlation between the pandemic and discretionary indicators supports the inclusion of interaction terms without introducing instability in the regression estimates.

III. Research Methodology

The primary empirical strategy employed in this study is a difference-in-differences (DiD) framework designed to estimate the effect of the COVID-19 pandemic on household spending allocation across expenditure categories. The baseline specification models category-level spending as a function of a pandemic indicator, a discretionary spending indicator, and their interaction. The interaction term captures whether discretionary categories experienced differential changes during the pandemic relative to non-discretionary categories, which serves as the key parameter of interest.

The identification strategy relies on treating the COVID-19 pandemic as an exogenous shock affecting consumption opportunities and economic uncertainty. Under this framework, the validity of the DiD estimator depends on the parallel trends assumption—that in the absence of the pandemic, discretionary and non-discretionary spending categories would have followed similar trends over time. While the limited number of pre-treatment periods (with 2019 as the primary baseline) constrains formal statistical testing of this assumption, graphical evidence (Figure 4) suggests broadly similar pre-pandemic spending patterns across categories.

Nevertheless, this assumption cannot be fully verified given the data limitations, and results should therefore be interpreted with appropriate caution.

The use of expenditure categories as treatment and control groups assumes that, absent the pandemic, discretionary and non-discretionary categories would have evolved similarly over

time. While the graphical evidence in Figure 4 supports this assumption, it may be violated if underlying demand trends differ systematically across categories.

To improve interpretability and account for proportional changes, a log-linear specification is also estimated, allowing coefficients to be interpreted as approximate percentage effects. In addition, a fixed-effects model is implemented to control for time-invariant differences across expenditure categories, such as structural consumption patterns or category-specific demand characteristics. This specification helps mitigate omitted variable bias arising from unobserved heterogeneity that may otherwise confound the estimated treatment effects.

A recovery specification is further introduced by incorporating a recovery indicator for 2023 and its interaction with discretionary spending. This extension allows the analysis to assess whether pandemic-induced changes in spending allocation persisted into the post-pandemic period.

All models are estimated using heteroskedasticity-robust standard errors to account for potential non-constant variance across expenditure categories. While the relatively small sample size limits formal diagnostic testing, robust standard errors help mitigate concerns about heteroskedasticity and improve the reliability of statistical inference. In particular, aggregating spending data may obscure heterogeneity across households and introduce measurement limitations that affect both coefficient estimates and standard errors.

Overall, while the DiD framework provides a useful approach for isolating differential changes across spending categories, the results should be interpreted as evidence of association consistent with causal mechanisms, rather than definitive causal estimates.

IV. Results

The main regression results are presented in Table 4, which reports estimates from the baseline, difference-in-differences, log-linear, and fixed-effects specifications. Across models, the coefficient on the pandemic indicator is generally not statistically significant, suggesting that aggregate household spending did not experience a large, uniform decline during the pandemic. This finding is consistent with existing evidence that fiscal transfers and forced savings mechanisms helped stabilize overall consumption levels.

In contrast, there is more consistent evidence of differential effects across spending categories. The coefficient on the discretionary indicator is negative and statistically significant across multiple specifications, indicating that discretionary categories tend to exhibit lower spending levels than essential categories. More importantly, the interaction between the pandemic and discretionary indicators is negative across all models and is statistically significant in the fixed-effects specification reported in Table 4. The estimated coefficient of approximately -558 implies that, on average, discretionary spending declined by roughly \$558 more per category relative to essential spending during the pandemic period. This provides economically meaningful evidence that discretionary spending contracted disproportionately, consistent with a reallocation of consumption toward essential goods and services.

However, the magnitude and statistical significance of this interaction vary across specifications, suggesting that the estimated effect's strength is sensitive to model choice. As such, the results should be interpreted as indicative of a compositional shift rather than precise estimates of effect size. The relatively small sample size may also contribute to large standard errors, limiting statistical power and reducing the likelihood of detecting significant effects.

The log-linear specification further supports this interpretation by indicating percentage reductions in discretionary spending during the pandemic. However, the interaction term is not statistically significant in that model. This highlights some uncertainty in the exact magnitude of the effect and reinforces the importance of cautious interpretation.

The recovery analysis presented in Table 5 suggests that while aggregate spending increased in the post-pandemic period, the estimated recovery coefficient is not statistically significant, indicating that the observed increase may not be distinguishable from random variation in the sample. Additionally, the interaction between recovery and discretionary spending remains negative, suggesting that discretionary categories did not fully converge back to pre-pandemic relative levels. This provides suggestive, but not conclusive, evidence of persistence in consumption reallocation.

Figures 2 and 3 visually reinforce these findings by illustrating a divergence between discretionary and non-discretionary spending during the pandemic, followed by only partial convergence in the recovery period. While the figures indicate a rebound in discretionary spending by 2023, Figure 3 shows that discretionary spending increases modestly relative to 2019, whereas essential spending increases substantially. This suggests that although discretionary categories recovered in absolute terms, the magnitude of recovery remains smaller than that observed for essential categories, reinforcing the conclusion of incomplete reallocation reversal.

The fixed-effects specification exhibits a very high R-squared value (0.997), which warrants careful interpretation. This high explanatory power primarily reflects the inclusion of category-specific fixed effects, which absorb nearly all time-invariant variation across expenditure categories. Given that spending levels differ substantially across categories (e.g., housing versus

entertainment), these fixed effects capture a large portion of total variation in the dependent variable. As a result, the R-squared in this context should not be interpreted as evidence of strong predictive performance, but rather as an indication that persistent differences across categories account for most of the variation. The within R-squared (0.186) provides a more informative measure of model fit, as it reflects the model's ability to explain variation over time within categories.

Overall, the results suggest that the pandemic did not lead to a uniform contraction in total spending, but instead induced a reallocation away from discretionary consumption toward essential categories. While this pattern is consistent across multiple specifications, the limited sample size, data aggregation, and sensitivity of the results to model choice imply that the conclusions should be interpreted as suggestive rather than definitive. These results rely on the parallel trends assumption discussed in the methodology section, which cannot be fully verified given the limited pre-pandemic period. They should therefore be interpreted with appropriate caution.

These findings are consistent with theoretical models of precautionary savings, such as Carroll (1997), which predict that households reduce non-essential expenditures in response to increased uncertainty while maintaining essential consumption.

An alternative interpretation of these findings is that the observed reallocation may reflect structural shifts in consumption opportunities rather than purely precautionary behaviour. For example, restrictions on services such as travel and dining may mechanically reduce discretionary spending, independent of household risk preferences. As such, the estimated effects likely capture a combination of behavioural responses and constraint-driven adjustments.

V. Conclusion

This paper examines how the COVID-19 pandemic affected household spending allocation using a difference-in-differences framework applied to Canadian household expenditure data. The results indicate that while total spending remained relatively stable, there was a significant reallocation away from discretionary spending categories during the pandemic, with only partial recovery in the post-pandemic period.

More broadly, these findings suggest that macroeconomic shocks such as the COVID-19 pandemic may induce persistent changes in consumption composition rather than temporary fluctuations in aggregate spending. This has important implications for both policymakers and investors, as sectors reliant on discretionary demand may experience longer-lasting shifts in consumption patterns. Understanding these compositional adjustments is therefore critical for anticipating the distributional and financial consequences of future economic shocks.

VI. Limitations and Future Research

Several limitations should be acknowledged. The relatively small sample size limits statistical power and may reduce the generalizability of the findings. The use of aggregated category-level data introduces potential aggregation bias and prevents analysis of heterogeneity across households. Additionally, the limited time horizon restricts the ability to assess long-term persistence in consumption behaviour.

Future research could address these limitations by incorporating micro-level household panel data, allowing for a more detailed analysis of heterogeneity across income and wealth groups. Further work could also explore the relationship between spending behaviour and household balance sheets, including wealth, debt, and savings. Finally, examining the role of credit access and liquidity constraints would provide deeper insight into the mechanisms underlying consumption adjustments.

References

- Andersen, A.L., Hansen, E.T., Johannesen, N. and Sheridan, A. (2022), Consumer responses to the COVID-19 crisis: evidence from bank account transaction data*. *Scandinavian Journal of Economics*, 124(3), 905-929. <https://doi.org/10.1111/sjoe.12512>
- Baker, S. R., Farrokhnia, R. A., Meyer, S., Pagel, M., Yannelis, C. (2020) How Does Household Spending Respond to an Epidemic? Consumption during the 2020 COVID-19 Pandemic, *The Review of Asset Pricing Studies*, 10(4), 834–862.
<https://doi.org/10.1093/rapstu/raaa009>
- Carroll, C. D. (1997). Buffer-Stock Saving and the Life Cycle/Permanent Income Hypothesis. *The Quarterly Journal of Economics*, 112(1), 1–55.
<https://doi.org/10.1162/003355397555109>
- Chetty, R., Friedman, J. N., Hendren, N., Stepner, M., & The Opportunity, I. T. (2020). The economic impacts of COVID-19: Evidence from a new public database built using private sector data (*NBER Working Paper No. 27431*) <https://doi.org/10.3386/w27431>
- Coibion, O., Gorodnichenko, Y., & Weber, M. (2020). The cost of the Covid-19 crisis: lockdowns, macroeconomic expectations, and consumer spending (*NBER Working Paper No. 27141*). <https://doi.org/10.3386/w27141>
- Gormsen, N. J., Koijen, R. S. J. (2020) Coronavirus: Impact on stock prices and growth expectations, *The Review of Asset Pricing Studies*, 10(4), 574–597
<https://doi.org/10.1093/rapstu/raaa013>

Kaplan, G., Moll, B., & Violante, G. L. (2020). The great lockdown and the big stimulus:

Tracing the pandemic possibility frontier for the U.S (*NBER Working Paper No.27794*).

<https://doi.org/10.3386/w27794>

Parker, J. A., Schild, J., Erhard, L., & Johnson, D. (2022). Household spending responses to the

economic impact payments of 2020: Evidence from the consumer expenditure survey

(*NBER Working Paper No. 29648*). <https://doi.org/10.3386/w29648>

Appendix A: Tables

Table 1 – Summary Statistics

Variable	Mean	SD	Min	Max
Total_Expenditure	98149.67	8746.93	92500	108225
Discretionary_Spending	36083.67	3154.43	32623	38798
NonDiscretionary_Spending	91488	9749.00	83962	102501
Pandemic	0.33	0.58	0	1
Recovery	0.33	0.58	0	1

Table 2 – Variable Definitions

Variable	Definition
spending	Average expenditure per household for a given category-year observation.
log_spending	Natural logarithm of spending.
pct_change_2019	Percent change in spending relative to the 2019 value within each category.
dollar_change_2019	Dollar change in spending relative to the 2019 value within each category.
pandemic	Indicator equal to 1 for 2021 observations and 0 otherwise.
recovery	Indicator equal to 1 for 2023 observations and 0 otherwise.
post	Indicator equal to 1 for post-2019 observations (2021 and 2023) and 0 for 2019.
discretionary	Indicator equal to 1 if the expenditure category is classified as discretionary and 0 if essential.
spend_type	Text label identifying each category as Essential or Discretionary.
base_2019	The 2019 spending level for a given expenditure category is used as the reference value.

Table 3 – Correlation Matrix

	spending	log_spending	pct_change_2019	dollar_change_2019	pandemic	recovery	post	discretionary
spending	1	0.813	0.039	0.591	-0.031	0.06	0.0	-0.455

							3	
log_spending	0.813	1	-0.039	0.378	-0.027	0.055	0.028	-0.543
pct_change_2019	0.039	-0.039	1	0.489	-0.079	0.316	0.237	0.014
dollar_change_2019	0.591	0.378	0.489	1	-0.168	0.333	0.164	-0.318
pandemic	-0.031	-0.027	-0.079	-0.168	1	-0.5	0.5	0
recovery	0.06	0.055	0.316	0.333	-0.5	1	0.5	0
post	0.03	0.028	0.237	0.164	0.5	0.5	1	0
discretionary	-0.455	-0.543	0.014	-0.318	0	0	0	1

Table 4 – Main Regression Results

	(1) Baseline	(2) DiD	(3) DiD Log	(4) Fixed Effects
Pandemic	-5.600	310.692	0.028	310.692**
	(1298.139)	(2480.032)	(0.351)	(126.954)
Discretionary		-4292.145**	-1.263***	
		(1816.889)	(0.370)	
Pandemic × Discretionary		-558.163	-0.046	-558.163**
		(2616.022)	(0.511)	(209.565)
Observations	60	60	60	60
R-squared	0.000	0.211	0.301	0.997
Within R-squared				0.186
RMSE	4943.17	4391.43	0.97	289.26
* p < 0.1, ** p < 0.05, *** p < 0.01				
Notes: Robust standard errors are reported in parentheses. The omitted year is 2019.				

Table 5 – Robustness Checks

	(1) Recovery	(2) Percent Change
Pandemic	310.692	-13.360**
	(2480.032)	(5.769)
Recovery	1426.077	
	(2794.012)	
Discretionary	-4292.145**	-0.571
	(1816.889)	(6.918)
Pandemic × Discretionary	-558.163	
	(2616.022)	
Recovery × Discretionary	-1310.312	
	(2926.514)	
Observations	90	60
R-squared	0.213	0.056
RMSE	4744.75	24.21
* p < 0.1, ** p < 0.05, *** p < 0.01		
Notes: Robust standard errors are reported in parentheses. The omitted year is 2019.		

Appendix B: Figures

Figure 1. Total Spending Over Time

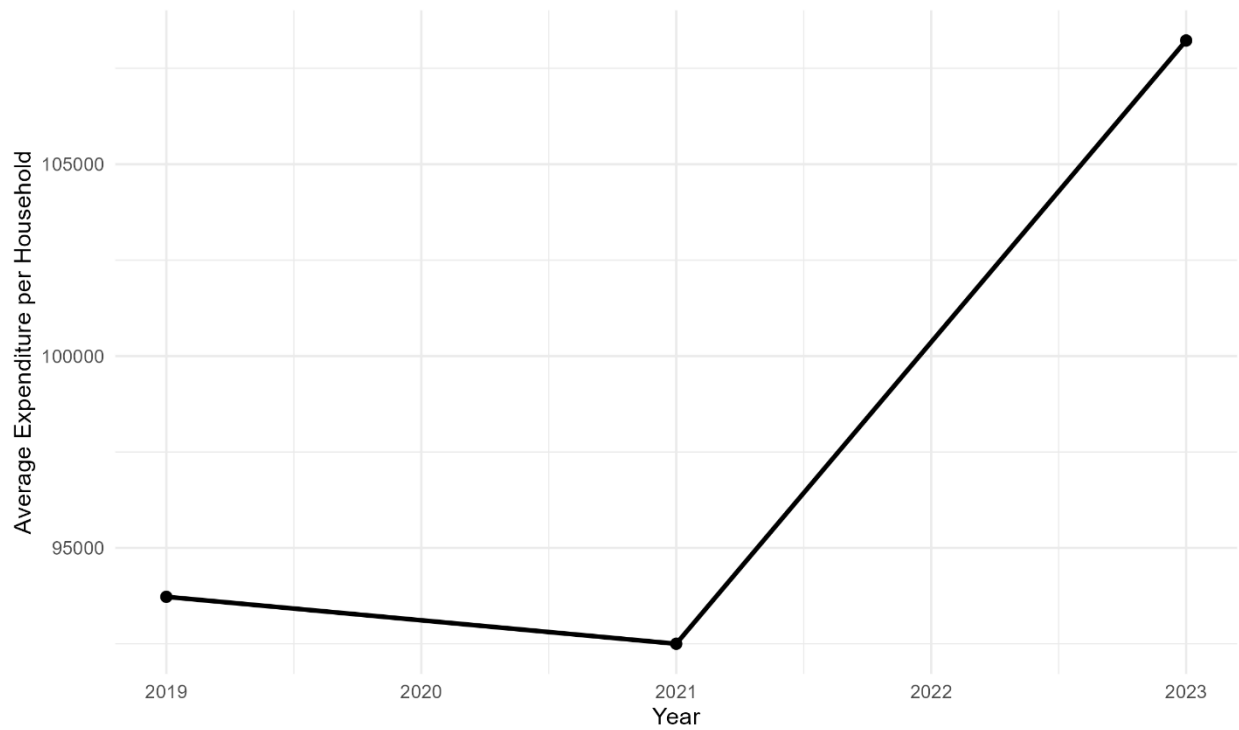


Figure 2. Discretionary vs Non-Discretionary Spending

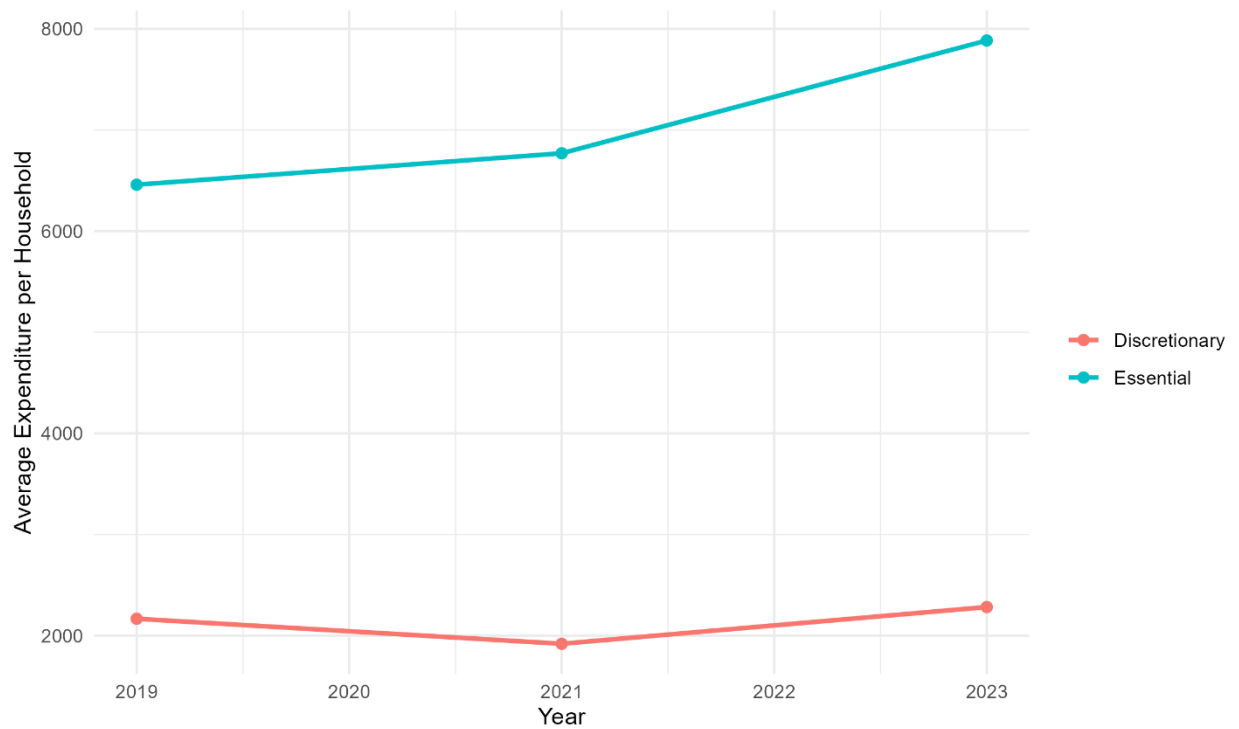


Figure 3. Percentage Change Relative to 2019

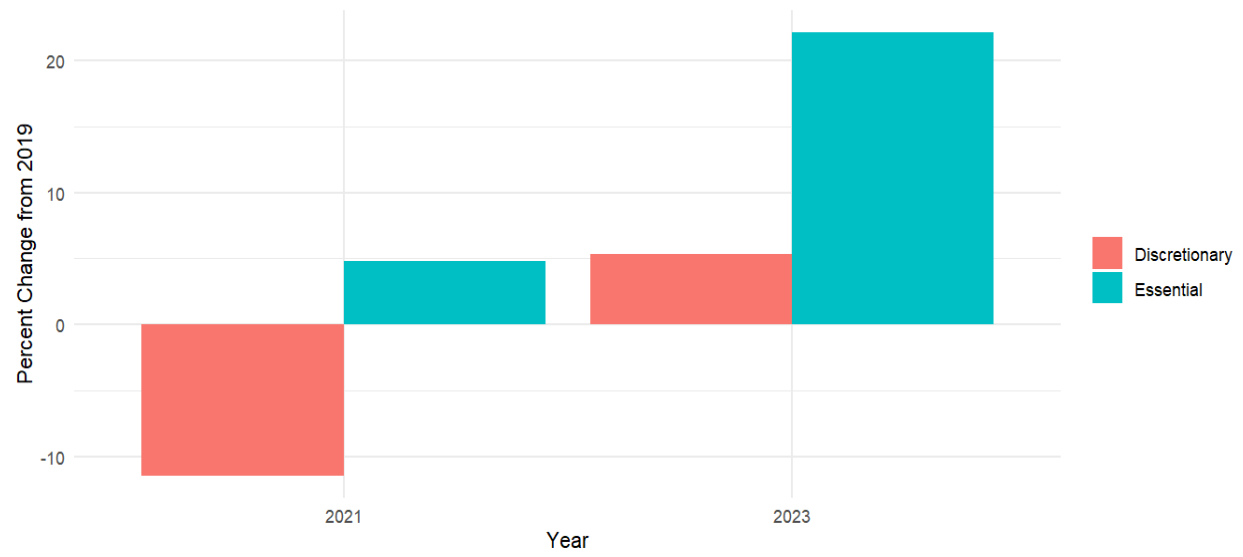


Figure 4. Difference-in-Differences Visualization

